# Machine Learning

## Problem Set 03

**Instructions.** If you think any question is too difficult, you can give up strategically. Please arrange your time according to your own schedule.

$\mathbb{1}\{\cdot\}$ represents the indicator function.

### Question 1.

Let $X = \mathbb{R}^2$, $Y \in \{0, 1\}$, and let $\mathcal{H}$ be the class of concentric circles in the plane, that is,

$$\mathcal{H} = \{h_r : r \in \mathbb{R}_+\}, \qquad \text{where} \qquad h_r(x) = \mathbb{1}\{\|x\| \leqslant r\}. \tag{1}$$

Prove that $\mathcal{H}$ is PAC learnable (assume realizability), and its sample complexity is bounded by

$$m_{\mathcal{H}}(\epsilon, \delta) \leqslant \left\lceil \frac{\log(1/\delta)}{\epsilon} \right\rceil. \tag{2}$$

### Question 2.

Compute the VC dimension of the following hypothesis functions.

**(a)** $h(x) = \mathbb{1}\{a < x\}$.

**(b)** $h(x) = \mathbb{1}\{a < x < b\}$.

**(c)** $h(x) = \mathbb{1}\{a \sin x > 0\}$.

**(d)** $h(x) = \mathbb{1}\{\sin(x + a) > 0\}$.

**(e)** $h(x) = \mathbb{1}\{\sin(ax) > 0\}$.

### Question 3.

Solve Exercise 7 in Chapter 3 of the book *Understanding Machine Learning: From Theory to Algorithms*:

**The Bayes optimal predictor:** Show that for every probability distribution $\mathcal{D}$, the Bayes optimal predictor $f_{\mathcal{D}}$ is optimal, in the sense that for every classifier $g$ from $\mathcal{X}$ to $\{0, 1\}$, $L_{\mathcal{D}}(f_{\mathcal{D}}) \leqslant L_{\mathcal{D}}(g)$.

### Question 4.

Solve Exercise 1 in Chapter 6 of the book *Understanding Machine Learning: From Theory to Algorithms*:

Show the following monotonicity property of VC-dimension: For every two hypothesis classes if $\mathcal{H}' \subseteq \mathcal{H}$, then $\text{VCdim}(\mathcal{H}') \leqslant \text{VCdim}(\mathcal{H})$.

## Question 5.

Recall that in class, we discussed incoherence. We say that $A \in \mathbb{R}^{n\times m}$ satisfy the **standard incoherence** condition with parameter $u$ if for rank-$r$ SVD of $A$ is $A = U\Sigma V^\top$, and for $i \in [n], j \in [m]$,

$$\left\|e_i^\top U\right\|_2 \leqslant \sqrt{\frac{ur}{n}}, \qquad \left\|e_j^\top V\right\|_2 \leqslant \sqrt{\frac{ur}{m}}, \tag{3}$$

for $u$ (therefore, not coherent with any basis). Here $U \in \mathbb{R}^{n\times r}, V \in \mathbb{R}^{m\times r}$, and $e_i$ are the $i$-th standard basis with appropriate dimension.

Given the following matrices, please compute the smallest value of $u$ that makes them satisfy the standard incoherence condition, respectively.

$$A_1 = \begin{bmatrix} 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 3 & 0 & 3 & 0 \\ 3 & 0 & 3 & 0 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 9 & 0 & 1 & 0 \\ 9 & 0 & 1 & 0 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \end{bmatrix}, \qquad A_3 = \begin{bmatrix} 9 & 1 & 2 & 1 \\ 9 & 1 & 2 & 1 \\ 2 & 1 & 2 & 3 \\ 2 & 1 & 2 & 3 \end{bmatrix}. \tag{4}$$